Week 10 web page: delivery guide

Instructor notes for Week10_ModelComparison.html

General

- Page: https://kevintshoemaker.github.io/NRES-746/Week10_ModelComparison.html. Open it before class; the sliders need no internet once loaded.
- The page shows code and output already run. For live demos, run the matching section of `CODE_Week10_ModelComparison.R` instead (same seeds, same numbers).
- Zoom the browser to 125-150% for the projector; the sliders scale with it.
- Questions on the page are printed without answers. Answers are below; the *Land:* line is where discussion should end up.
- Use the floating table of contents to jump; it doubles as the outline.

Part 1: LRT, AIC, regularization (Monday, ~35 min)

Big picture (~4 min)

- Point at COMPARE in the workflow figure, then the three uses (null tests, selection, multimodel inference).
- Order to stress: validation and CV are the gold standard; LRT and AIC are single-fit shortcuts; ridge and lasso are a third route.

WOD: bias-variance tradeoff (~8 min)

- Show the WOD figure first; ask a few students for their 19% predictions.
- Then scroll to the 20-sample figure. Point at the right panel: both sets center near the truth (red), but the curve's spread is about 2.4 times the line's (SD 2.9 vs. 1.2 cm).
- If asked about bias: in this simulation the truth is a line, so the line is unbiased too. A constant model would show bias.

Likelihood ratio test (~8 min)

- Walk the four numbered steps, then the code: $D = 2.01$ vs. cutoff 7.81, $p = 0.57$.
- "Is the Poisson model nested in the negative binomial? Can we use an LRT here?" *Land:* yes, nested ($k \to \infty$), but that value is on the boundary of the parameter space, so the naive p is about twice too big. The deviance here is in the thousands, so it doesn't matter.

AIC (~10 min)

- "The no-wave model has weight 0.52. Is there a 52% chance it's the true model?" *Land:* no; relative support within this set (Bolker p. 284).
- `const_ak` is 1.48 AIC units behind with one useless parameter: "within 2" is not support (Arnold 2010).

Regularization (~12 min)

- Say up front the creeper data are simulated, so we know the truth.
- "Which covariates show up in nearly every top model?" *Land:* big trees, snags, road distance; the others are the best model plus one useless covariate, each about 2 units behind.
- Path plots: read the lasso right to left; big trees and snags enter first, then road distance, well before the rest.
- "The lasso estimates road distance at 0.07; the truth is 0.3. Is the lasso broken?" *Land:* no. Shrinkage trades bias for lower variance and better prediction. Use the lasso for prediction and covariate choice, not effect sizes (refit, or a shrinkage prior).
- 40 simulated data sets: AIC found the exact true set 12 times, lasso 13.
- **Behind?** Stop after the path plots; the CV demo can open the next lecture.

Discussion (~5 min)

- Q1 (northness) *Land:* report the uncertainty, not one list; big trees and snags are the safest claims.
- Q2 (grazing at 0) *Land:* zero means "not needed to predict counts here," not "no effect." Grazing may act through shrub cover or be too small to detect with 50 stations. A management question is causal: COMPARE part 3.

Part 2: budgets and CV (Lab 8 pre-lab)

Lasso slider (`lasso_kink.html`)

- Start at $\lambda = 0$: dot at the MLE (0.4), both estimates equal.
- Move to 0.4: lasso estimate 0.2. Then to 1 and point at ridge: also 0.2. Same budget $t = 0.2$, different λ.
- Ask where the lasso dot will land at 0.8 before moving there. It hits the kink at 0 and stays for every larger λ (try 1.5).
- Ridge keeps sinking but never reaches 0 (0.16 at $\lambda = 1.5$).
- "Why does the lasso stop at exactly 0 once $\lambda \geq 0.8$?" *Land:* at $\beta = 0$ the NLL slopes downhill at rate $2(0.4) = 0.8$. The lasso charges λ per unit of $|\beta|$ from the first unit, so once $\lambda \geq 0.8$, leaving 0 costs more than it gains. Ridge's penalty is flat at 0 (slope $2\lambda\beta = 0$), so the first bit of coefficient is free.
- Don't draw the 2-D picture; Act 1 Part 2 asks for it.
- Glmnet tips and CV folds: read off the page; don't give away why standardize or whether to penalize β_0 (Act 2 1a-1b).

Part 3: Bayesian comparison (Wednesday, ~35 min)

Why not AIC? (~3 min) No single $\hat{\theta}$; priors blur the parameter count.

Ridge is a prior (~6 min)

- Do the six steps on the board; the page has them for reference.
- Ridge-prior slider (`ridge_prior.html`):
  - Default $\tau = 0.05$: $\lambda = 200$, prior worth 9 routes at 0, shrinkage factor 0.47, estimate 0.031 (MLE 0.065).
  - Slide to 0.15: $\lambda = 22$, prior worth 1 route, factor 0.89, estimate 0.058. The posterior moves toward the likelihood.
  - Slide to 0.01: the prior dominates; estimate near 0.
  - Point out the posterior is narrower than both the prior and the likelihood (precisions add).
- "Which prior matches the lasso penalty?" *Land:* Laplace (double exponential), $\propto e^{-\lambda|\mu|}$: a spike at 0 and heavier tails.

WAIC and LOO (~12 min)

- Stan chunks take about a minute to run live; run them before class.
- "The full model is 2.1 elpd behind, SE 1.8. What can you conclude?" *Land:* about one SE; the data can't tell the models apart. Not the same as "no wave effect."
- **Behind?** Skip the DIC row.

Shrinkage priors (~12 min)

- Interval plot: northness (truth 0) under each prior; triangles are the glmnet lasso.
- "The lasso set northness to 0; the Laplace posterior says 0.08 (-0.07 to 0.28). Which is more honest?" *Land:* the posterior keeps the uncertainty a single zero hides; a Bayesian lasso never gives exact zeros.
- Horseshoe: sketch only.

Discussion (~5 min)

- Q1 (wave "no effect") *Land:* within about 1 SE is inconclusive; "not detected" is not "no effect"; corroborated, not proven.
- Q2 (northness) *Land:* for prediction either works; for inference, report the interval.
- Bonus (Bayes factors, RJMCMC): only if time remains.